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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M. Tech (CCT) (CBCS) Degree Examinations

May – June 2025

22CCT223 – BUILDING SERVICES & MAINTENANCE

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each
and **One full** question from **Unit – III**.

		Unit – I	Marks	BT*	CO*	PO*
1.	a)	What are the key benefits of incorporating fire resistance into building design?	10	L*1	1	1
	b)	Explain the key principles for designing effective means of escape in buildings.	10	L3	1	1, 5
2.	a)	Explain the purpose and objectives of the NBC in promoting fire safety in buildings	10	L2	2	1, 2, 5
	b)	Why is it necessary to classify buildings based on fire load, and how is this classification carried out?	10	L3	2	1, 2, 5
3.	a)	Evaluate the pros and cons of centralized vs. decentralized engineering service systems for different building types	10	L5	3	1
	b)	Evaluate the environmental impact of lifts and escalators. Are there alternative methods for vertical transportation with a lower impact?	10	L5	3	1
		Unit – II				
4.	a)	Identify potential problems that could cause low water pressure in a building's cold-water system.	10	L4	4	1, 2, 5
	b)	How can you educate others in your building or community about best practices to clog free working of waste water system?	10	L3	4	1, 2, 5
5.	a)	How can building management encourage residents to reduce, reuse, and recycle solid waste?	10	L3	4	1, 2, 5
	b)	Analyze the factors to be considered for installing solar panels on the roof of a specific building.	10	L4	4	1, 2, 5
6.	a)	How can an organization ensure effective maintenance management, and what components are crucial?	10	L3	4	1, 2, 5
	b)	Identify the potential root causes behind a recurring plumbing leak in a building.	10	L3	4	1, 2, 5
		Unit – III				
7.	a)	Analyse the factors that contribute to the deterioration of building materials over time.	10	L4	5	1, 5
	b)	Investigate the causes of foundation settlement and propose preventive measures.	10	L4	5	1, 5

P.T.O.

8. a) Analyse the benefits and drawbacks of using epoxy to repair a cracked tile floor.
b) Explain the purpose of waterproofing a basement.

10	L4	5	1, 5
10	L2	5	1, 5

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
